

PDE in distributional sense

$$u' = u_0$$

$\phi \in C^\infty$ with compact support

$$-\langle u, \phi' \rangle = - \int u \phi' = \int u' \phi = \langle u', \phi \rangle = \langle u_0, \phi \rangle$$

GP with symmetry constraint

GP with odd sample paths. For a function,

$$f \text{ odd if } f(-x) = -f(x)$$

$$Y(x) = Y_0(x) - Y_0(-x)$$

$$Y = T(Y_0) \text{ with } Tf = x \mapsto f(x) - f(-x)$$

$$Y(-x) = Y_0(-x) - Y_0(-(-x)) = -Y(x)$$

Let us choose $Y_0 \sim GP(0, k_0)$. We can prove that $Y \sim GP(0, k)$ and we can compute k :

$$\begin{aligned} k_Y(s, t) &= \text{cov}(Y(s), Y(t)) \\ &= \text{cov}(Y_0(s) - Y_0(-s), Y_0(t) - Y_0(-t)) \\ &= \text{cov}(Y_0(s), Y_0(t)) - \text{cov}(Y_0(s), Y_0(-t)) - \text{cov}(Y_0(-s), Y_0(t)) + \text{cov}(Y_0(-s), Y_0(-t)) \\ &= k_0(s, t) - k_0(s, -t) - k_0(-s, t) + k_0(-s, -t) \end{aligned}$$

GP for heat equation

$$\begin{aligned} \text{Cov}(u(x, t), u(x', t')) &= \text{Cov} \left(\int G(x - y, t) \phi(y) dy, \int G(x' - y', t') \phi(y') dy' \right) \\ &= \iint \text{Cov}(G(x - y, t) \phi(y), G(x' - y', t') \phi(y')) dy dy' \\ &= \iint G(x - y, t) G(x' - y', t') \text{Cov}(\phi(y), \phi(y')) dy dy' \\ &= \iint G(x - y, t) G(x' - y', t') k_\phi(y, y') dy dy' \end{aligned}$$

Recall that

$$G \otimes G(A, B) = G(A)G(B)$$

Thus, we have the compact writing

$$\begin{aligned} \text{Cov}(u(x, t), u(x', t')) &= \iint G \otimes G((x - y, x' - y'), (t, t')) k_\phi(y, y') dy dy' \\ &= [(G \otimes G) * k_\phi]((x, t), (x', t')) \end{aligned}$$

and thus

$$k_u = (G \otimes G) * k_\phi$$

RKHS

$$\langle k(x, \cdot), \partial_x k(x', \cdot) \rangle_H = \partial_y \langle k(x, \cdot), k(x', \cdot) \rangle_H = \partial_y k(x, x')$$

$$Cov(Y(s), Y'(t)) = \partial_y Cov(Y(s), Y(t)) = \partial_y k_Y(s, t)$$

$$Cov(Y'(s), Y'(t)) = \partial_x \partial_y Cov(Y(s), Y(t)) = \partial_{xy}^2 k_Y(s, t)$$

Notebooks

$$k(x, y) = \sigma^2 \exp \left(- \left[\frac{x - y}{\ell} \right]^2 \right) = \exp \left(-\theta [x - y]^2 \right)$$